RFID Logging System

Presented by

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CMPE-246 Final Project

Project Overview

This project presents an RFID-based system that automates access control and attendance tracking. Users scan RFID cards, and the system validates access while recording attendance data.

Murat worked on backend development and system integration, while Adam focused on hardware setup and implementation.

The project was developed through stages including design, prototyping, integration, and testing.

Objective and Scope

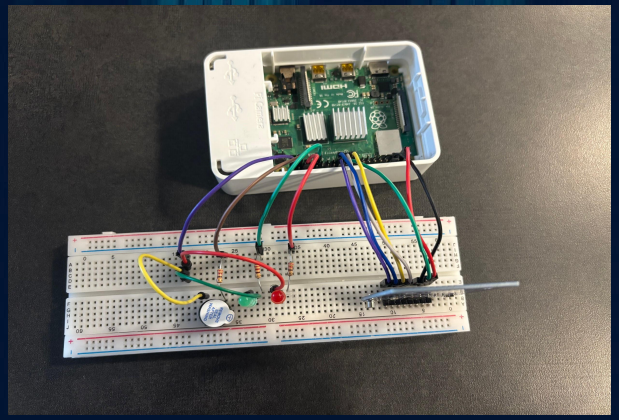
This project aims to automate both access control and attendance tracking using RFID and provide surveillance.

The system uses a Raspberry Pi connected to an RFID reader, which communicates with a Flask backend and SQLite database.

The scope includes validation, attendance logging, and a web-based dashboard. It’s also built for modularity as there a lot of possible improvements that can be done

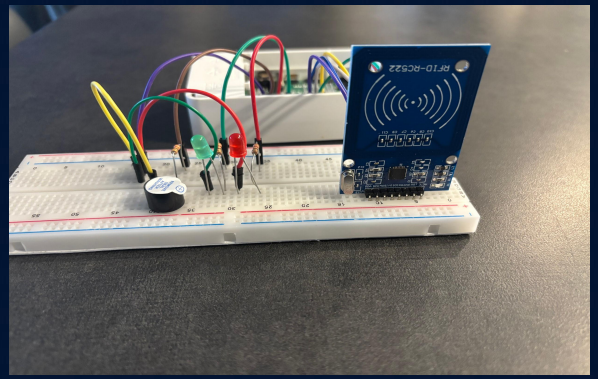
Hardware

Hardware system built around Raspberry Pi 4 as a controller. LED, Buzzers, and RFID Module connected through the GPIO interface. RFID module communicates with the Pi using SPI protocol.



When a card is scanned, the reader captures the UID and sends it to the system for validation.

For output feedback, a buzzer and LEDs were used indicating a successful or failed scan with different colors and audio output.



Software

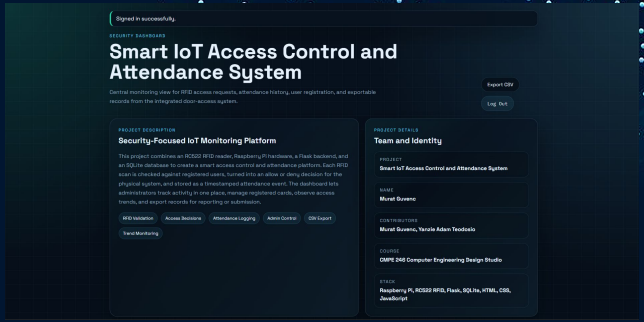
The software system is built using Python with a Flask-based backend that handles communication between the hardware and database.

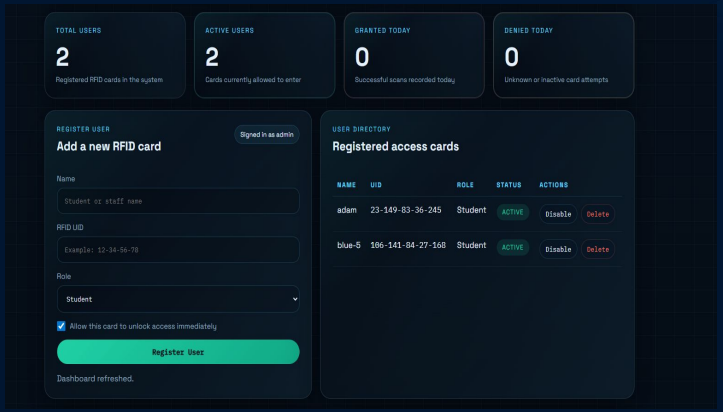
When an RFID card is scanned, the Raspberry Pi sends the UID to the backend through an HTTP request. The server validates the UID and determines whether access is granted or denied.

An SQLite database is used to store registered users and attendance records, including timestamps and scan status.

A web-based dashboard allows administrators to view attendance logs, manage users, and export data as CSV files.

The system also includes logic to prevent duplicate scans, ensuring accurate attendance tracking.





Results and Findings

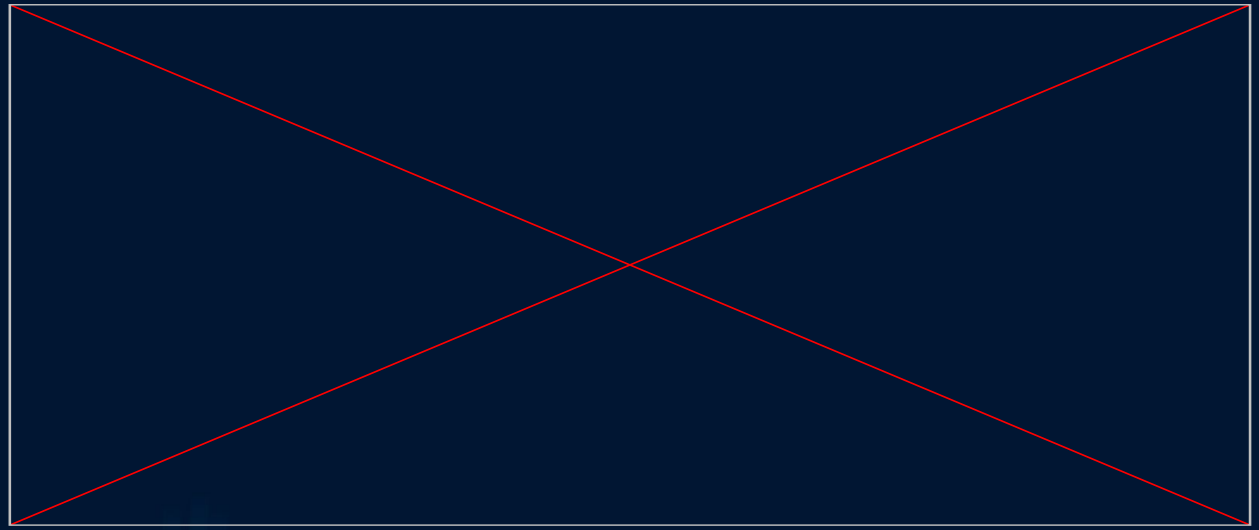
The system successfully works end-to-end, from scanning an RFID card to validating it through the backend and recording attendance.

Each scan sends a request to the server, which verifies the user and stores a timestamped record in the database.

The dashboard displays attendance logs and summary information in real time.
CSV export functionality allows data to be downloaded for external use.

The system also handles duplicate scans to prevent incorrect multiple entries.

Demonstratio n



Reflection and Conclusions

Integrating hardware and software components was one of the main challenges in this project.
Ensuring reliable communication between the RFID reader, Raspberry Pi, and backend required careful debugging.

Debugged the library and source code for the RFID module due to incompatible imports and syntax errors during early hardware testing.

Duplicate scan handling was an important issue discovered during testing and required additional logic to resolve.

Through this project, we gained experience in Flask backend development, SQLite databases, and Raspberry Pi GPIO communication.

Overall, the project demonstrates a complete IoT system from hardware interaction to data visualization.

References

This project used several key resources for both hardware and software development.

We referred to official documentation for Raspberry Pi, Flask, and SQLite, along with the MFRC522 RFID library.

Additional support came from online tutorials, guides, and community resources that helped with implementation and troubleshooting.

Public Dissemination

The project is publicly available on GitHub, where all source code and documentation are stored.

The repository includes backend code, hardware integration scripts, and setup instructions.

GitHub was used for version control and collaboration throughout development.

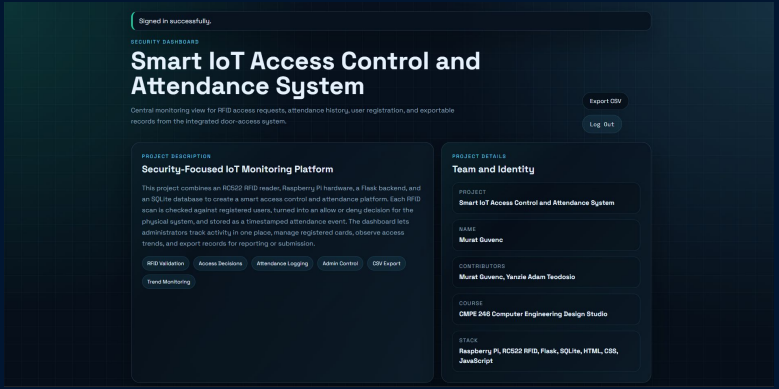
It also allows others to review the project, test it, and provide feedback for improvement.

README

CMPE 246 — Smart IoT Access Control & Attendance System

An RFID-based access control system built with a Raspberry Pi and a Flask web dashboard. When a card is tapped on the RFID reader, the system checks it against a SQLite database, logs the event, and gives instant feedback via LEDs and a buzzer. Authorized users get a green light and two beeps; unauthorized cards get a red light and a long beep. All activity is visible in real time on the web dashboard.

System Overview

```
[ RFID Card ] --> [ RC522 Reader ]
                        |
              [ Raspberry Pi ]
              Backend_Hardware_Client.py
                        |  HTTP POST /api/scan
                        v
              [ Flask Server ]  <-->  [ SQLite Database ]
                        |
              [ Web Dashboard ]
              http://<server-ip>:5000
```

Hardware side: Raspberry Pi reads RFID cards and sends UIDs to the Flask backend over HTTP.
Software side: Flask server validates the UID, logs the attendance record, and returns granted/denied.